

Singular support

k field, X k -scheme of f.t. $\leadsto D(X) = \text{derived cat. of } \mathcal{O}_X\text{-sheaves on } X$
of char p $\mathcal{O}_X = \overline{\mathbb{F}_p}$ $l \neq p$
sheaf $= \text{ob}(D(X))$

Goal: X smooth/ k
 $F \in D(X) \leadsto SS(F) \subset T^*X$ closed conical subset.

which captures the "micro-local" aspect of F .

Closed conical subset

$\begin{matrix} E \\ \pi \downarrow \\ X \end{matrix}$ vector bundle, $C \subset E$ closed conical subset: a closed subset
stable under the fibrewise
 G_m -action.

A closed conical subset $C \subset E$ is determined by

— its base $B(C) \subset X$
 $\parallel$
 $\pi(C)$

— its projectivisation $\mathbb{P}(C) \subset \mathbb{P}(E)$
 $\parallel$
 $(C \setminus X) / G_m$

Def. X smooth/ k . $C \subset T^*X$ closed conical subset.

$h: W \rightarrow X$ morphism of smooth schemes/ k is C -transversal if

$\underbrace{h^*C}_{\parallel} \cap \ker(T^*X \times_x W \rightarrow T^*W) \subset 0\text{-section.}$

inverse image of C under $[T^*X \times_x W \rightarrow T^*X]$.

Ex. 1. h smooth $\Rightarrow$ C -transversal for any C .

2. $C \subset 0$ -section $\Rightarrow$ every h is C -transversal

$$\begin{array}{ccc}
 h: W \rightarrow X & C\text{-transversal} & \begin{array}{ccc}
 \xrightarrow{\text{image} =: h^0 C} & & \\
 h^* C \rightarrow T^* X \times_X W & \rightarrow & T^* W \\
 \downarrow & & \downarrow \\
 C & \rightarrow & T^* X
 \end{array}
 \end{array}$$

Claim. $h^0 C$ is a closed conical subset of $T^* W$

[can show that $h^* C \rightarrow T^* W$ is finite]

local computation.

Def. X smooth / k . $C \subset T^* X$ closed conical subset.

$f: X \rightarrow Y$ morphism of smooth schemes / k is C -acyclic if

$$\left(T^* Y \times_Y X \rightarrow T^* X \right)^{-1}(C) \subset 0\text{-section}.$$

Ex. 1. $C = T_X^* X$ 0-section, f smooth $\Rightarrow f$ is C -acyclic

2. $X \xrightarrow{f} \text{Spec } k$ is C -acyclic for every $C \subset T^* X$.

3. $X \xrightarrow{f} Y$ C -acyclic, $Y \xrightarrow{g} Z$ smooth $\Rightarrow gf: X \rightarrow Z$ is C -acyclic

4. $\text{id}_X: X \rightarrow X$ is C -acyclic $\Leftrightarrow C \subset T_X^* X$.

Lemma. f C -acyclic $\Rightarrow f$ is smooth over a nbhd of $B(C)$.

Local acyclicity $f: X \rightarrow Y$ morphism of schemes / k , $F \in D(X)$

f is F -acyclic if any specialisation $t \rightsquigarrow s$ of geom. pts of Y ,

$$\begin{array}{ccccc}
 X_s & \xrightarrow{i} & X \times_Y Y(s) & \xleftarrow{j} & X_t \\
 \downarrow & & \downarrow & & \downarrow \\
 s & \longrightarrow & Y(s) & \longleftarrow & t
 \end{array}$$

$$F_{X_s} \xrightarrow{\sim} i^* j^* F_{X_t}.$$

(Equivalently, $\forall$ geom. pt $x \rightarrow X_s$, $F_x \xrightarrow{\sim} R\Gamma(X_{(x)} \times_{Y(s)} t, F_{X_{(x)} \times_{Y(s)} t})$)
 $\uparrow$
 Milnor fibre.

f is universally F -acyclic if $\forall g: Y' \rightarrow Y$, $f': X' = X \times_Y Y' \rightarrow Y'$ is $g'^* F$ -acyclic.

$$\begin{array}{ccc}
 X' & \xrightarrow{f'} & Y' \\
 g' \downarrow & & \downarrow g \\
 X & \xrightarrow{f} & Y
 \end{array}$$

Ex. 1. $f: X \rightarrow Y$ smooth, F locally constant $\Rightarrow f$ is F -acyclic

(local acyclicity of smooth morphisms)

2. $f: X \rightarrow \text{Spec } k$, F -arbitrary, f is universally F -acyclic.

3. $f: X \rightarrow Y$ F -acyclic, $g: Y \rightarrow Z$ smooth $\Rightarrow g^* F$ -acyclic (generic universal local acyclicity)

4. $\text{id}_X: X \rightarrow X$ is F -acyclic $\Leftrightarrow F$ is locally constant (Illusie's appendix to [Théorèmes de finitude])

Def. X smooth $/k$, $C \subset T^*X$ closed conical subset, F sheaf on X .

We say that F is micro-supported on C if :

(h.t) $\left[\begin{array}{l} \forall h: W \rightarrow X, f: W \rightarrow Y \text{ morphisms of smooth schemes } /k, \\ C\text{-acyclic } h \text{ } C\text{-transversal, } f \cdot h^* C\text{-acyclic} \Rightarrow f \text{ is } h^* F\text{-acyclic.} \end{array} \right.$
 $\Rightarrow$ (h.t): $W \rightarrow X \times Y$ $C \times T^*Y$ -transversal (universally)

Ex. 9. 1. F micro supp. on C , f C -acyclic $\Rightarrow f$ F -acyclic.

2. Every sheaf on X is micro-supp. on T^*X .

(h, f) T^*X -acyclic $\Rightarrow (h, f): W \rightarrow X \times Y$ is smooth

Problem is étale local $\Rightarrow W = X \times Y \times \mathbb{A}^n$.

$$\begin{array}{ccccc} F_! & & \text{local} & & \\ & & h^*F & & \\ X & \leftarrow & X \times Y & \leftarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec } k & \leftarrow & Y & \leftarrow & Y \times \mathbb{A}^n \end{array}$$

generic local acyclicity:

$W \rightarrow Y \times \mathbb{A}^n$ is h^*F -acyclic

$Y \times \mathbb{A}^n \rightarrow Y$ is smooth

$\Rightarrow W \rightarrow Y$ is h^*F -acyclic

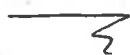
3. F micro supp. on $C \Rightarrow \text{Supp}(F) \subset B(C)$.

4. F locally constant $\Rightarrow F$ is micro supp. on the 0-section T_X^*X .

5. F micro supp. on C , $U \subset X$ open, $F|_U$ micro supp. on $C' \subset T^*U$
 $\Rightarrow F$ is micro supp. on $C|_{X-U} \cup \overline{C'}$.

Def The singular support $SS(F)$ of F is the smallest closed conical subset of T^*X on which F is micro-supported.

Th. (Beilinson): $\forall F \in D(X)$, $SS(F)$ exists.

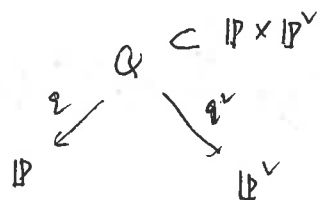


- Reduction to $\mathbb{P}^n$:
- ① reduction to affine schemes: can glue SS along an open cover
 - ② reduction to $\mathbb{A}^n$: $X \xrightarrow{i} P$ closed immersion, can reconstruct $SS(F)$ from $SS(i^*F)$
 - ③ reduction to $\mathbb{P}^n$: can restrict SS to an open subset.

Proof of the $\mathbb{P}^n$ case

V k -vec. sp. of dim $n+1$, $\mathbb{P} = \mathbb{P}(V)$, $\mathbb{P}^\vee = \mathbb{P}(V^\vee)$

Incidence correspondence



Radon transform

$$R = q_* q^* [n-1]: D(\mathbb{P}) \rightarrow D(\mathbb{P}^\vee)$$

$$R^\vee = q_* q^{\vee*} [n-1]: D(\mathbb{P}^\vee) \rightarrow D(\mathbb{P})$$

Th. (Brylinski)

R & $R^\vee(n-1)$ are both left & right adjoint to each other.

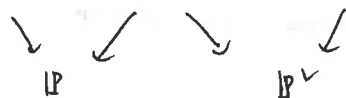
$\forall F \in D(\mathbb{P}) \simeq D(\mathbb{P}^\vee)$, $\text{fib}(RR^\vee(F)(n-1) \rightarrow F)$
 $\text{fib}(F \rightarrow R^\vee R(F)(n-1))$ locally constant.

Cor. F is micro supp. on $C \Rightarrow R^\vee R(F)$ micro supp. on $C^\perp = C \cup T_{\mathbb{P}}^* \mathbb{P}$.

Legendre transform

We have natural identifications

$$\mathbb{P}(T^* \mathbb{P}) \xleftarrow{\sim} Q \xrightarrow{\sim} \mathbb{P}(T^* \mathbb{P}^\vee)$$



Q has a tautological line bundle $\lambda: \forall (x, x^\vee) \in Q$, $\lambda(x, x^\vee) =$ the conormal

λ itself can be identified w/ the conormal bundle

line of $x^\vee$ at x .

If the closed immersion $Q \hookrightarrow \mathbb{P} \times \mathbb{P}^\vee$.

Remark Under the Legendre transform, λ induces the canonical contact structure on $\mathbb{P}(T^* \mathbb{P})$, $\mathbb{P}(T^* \mathbb{P}^\vee)$.

$g: X \rightarrow X'$ be a proper map between smooth schemes $/k$.

$C \subset T^*X$ closed conical subset of T^*X

$$g_* C := \underbrace{\left(T^*X' \times_{X'} X \xrightarrow{pr_2} T^*X' \right)}_{\text{proper}} \left(T^*X' \times_{X'} X \xrightarrow{dg} T^*X \right)^{-1} (C)$$

Prop X smooth $/k$, $F \in D(X)$, $C \subset T^*X$ closed conical subset

1. If $h: W \rightarrow X$ is C -transversal, then h^*F is micro supp. on h^*C

2. If $g: X \rightarrow X'$ is proper, then g_*F is micro-supp. on g_*C .

Def For $C \subset T^*\mathbb{P}$ closed conical subset,

$$C^\vee := q_0^\vee q^0 C \subset T^*\mathbb{P}^\vee$$

Legendre transform of C .

Cor. F micro supp. on C

$\Downarrow$

RF micro supp. on $C^\vee$

$\Downarrow$

F micro supp. on $C^{\vee\vee} +$

Prop Let $C \subset T^*\mathbb{P}$ be a closed conical subset. Let $E = \mathbb{P}(C) \subset \mathbb{P}(T^*\mathbb{P}) \simeq \mathbb{Q}$.

1. $E \subset \mathbb{Q}$ is the complement to the largest open subset where $(q, q^\vee)$ is C -acyclic.

$$2. \mathbb{P}(C^\vee) = \mathbb{P}(C)$$

$$3. C^{\vee\vee} \subset C^+$$

Proof 1. $(q, q^\vee): \mathbb{Q} \rightarrow \mathbb{P} \times \mathbb{P}^\vee$ is C -acyclic at $(x, x^\vee) \in \mathbb{Q}$

$$\Leftrightarrow \ker \left(dq_{(x, x^\vee)} \times dq_{(x, x^\vee)}^\vee : T_x^* \mathbb{P} \times T_{x^\vee}^* \mathbb{P}^\vee \rightarrow T_{(x, x^\vee)}^* \mathbb{Q} \right) \cap (C_x \times T_{x^\vee}^* \mathbb{P}^\vee) \subset \{0\}$$

$(q, q^\vee)$ not C -acyclic at $(x, x^\vee) \in Q$

$$\Leftrightarrow \exists \neq (p, p^\vee) \in (x \times T_{x^\vee}^* \mathbb{P}^\vee$$

$$\text{s.t. } dq_{(x, x^\vee)}(p) + dq^\vee_{(x, x^\vee)}(p^\vee) = 0$$

Since $dq^\vee_{(x, x^\vee)}$ injective, $p \neq 0$.

$$p^\vee \text{ determined by } dq^\vee(p^\vee) = -dq(p)$$

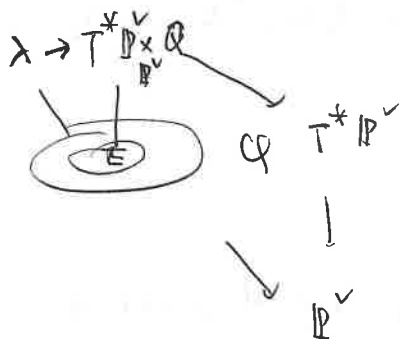
$$\Leftrightarrow (x, x^\vee) \in E = \mathbb{P}(C)$$

intersection inside T^*Q .

$$2. \text{ Recall the def'n of } C^\vee: q^*C \cap q^{\vee*}T^*\mathbb{P}^\vee \subset q^*T^*\mathbb{P} \cap q^{\vee*}T^*\mathbb{P}^\vee \simeq \lambda \subset T^*Q.$$

$$C^\vee = \text{image under } (q^{\vee*}T^*\mathbb{P}^\vee \rightarrow T^*\mathbb{P}^\vee)$$

$$\text{But } q^*C \cap q^{\vee*}T^*\mathbb{P}^\vee \simeq (C \times_{\mathbb{P}} Q) \cap \lambda \subset T^*Q$$



$$\simeq \lambda|_E \text{ up to base. } (C \text{ replaced by } C \setminus 0\text{-sec.})$$

$$\Rightarrow \mathbb{P}(C^\vee) = \mathbb{P}(C)$$

$$3. \mathbb{P}(C^{\vee\vee}) = \mathbb{P}(C^\vee) = \mathbb{P}(C) \Rightarrow C^{\vee\vee} \subset C \cup 0\text{-section.}$$

Prop. (1) F is micro supp. on C

(2) $q^\vee$ is universally q^*F -acyclic outside $E = \mathbb{P}(C)$

(3) RF is micro supp. on $C^{\vee+}$.

(4) F is micro supp. on C^+

then (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4)

Proof (1) $\Rightarrow$ (2) : $(q, q^\vee)$ is C -acyclic outside E .

(2) $\Rightarrow$ (3) Assume (h, t) is $C^{\vee+}$ -acyclic.

$$\begin{array}{ccccc}
 & E & \longleftarrow & E_W & \\
 & \downarrow & & \downarrow & \\
 \mathbb{P} & \xleftarrow{q} & Q & \xleftarrow{h'} & Q_W \\
 & \downarrow q^\vee & & \downarrow q_W^\vee & \searrow f' \\
 & \mathbb{P}^\vee & \xleftarrow{h} & W & \xrightarrow{t} Y
 \end{array}$$

WTS t is $\underbrace{h^* RF}_{\cong} \text{-acyclic}$
 $q_W^\vee (qh')^* F$

suffices to show that f' is $(qh')^* F$ -acyclic.

Outside E , $q^\vee$ is universally $q^* F$ -acyclic $\Rightarrow$ outside E_W ,

$q_W^\vee$ is $(qh')^* F$ -acyclic.

(h, t) is $T_{\mathbb{P}}^* \vee \mathbb{P}^\vee$ -acyclic $\Rightarrow$ t smooth $\Rightarrow$ f' is $(qh')^* F$ -acyclic.
 On a nbhd of E_W ,

(h, t) $C^\vee$ -acyclic $\Rightarrow$ (h', f') is $q^0 C$ -acyclic.

$q^\vee \circ q^0 C \Rightarrow (qh', f')$ is C -acyclic

Conormal bundle of $Q_W \rightarrow \mathbb{P} \times W$ = pullback of conormal bundle of

$$\underbrace{Q \rightarrow \mathbb{P} \times \mathbb{P}^\vee}_{\lambda}$$

$$(C \times T^* W) \cap \lambda_W \supset (T^* \mathbb{P} \times T^* W) \cap \lambda_W|_{E_W}$$

$f: W \rightarrow Y$ smooth. $\Rightarrow$

0-section $\supset (C \times T^* Y) \cap \text{conormal}$, $\supset (T^* \mathbb{P} \times T^* Y) \cap \text{conormal}(Q_W \rightarrow \mathbb{P} \times Y)|_E$
 $(Q_W \rightarrow \mathbb{P} \times Y)$ anbd

$\Rightarrow (qh', f')$ is $T^* \mathbb{P}$ -acyclic over U of E_W , F is micro supp on $T^* \mathbb{P}$

$\Rightarrow f'$ is $(qh')^*$ F -analytic on U . $\square$

(3) $\Rightarrow$ (4) By (3) & (1) $\Rightarrow$ (3), $R^* R F$ is micro supp. on $(C^+)^{v+} = C^+$.

By Radon transform, F is also micro supp. on C^+ .

Cor. $F \in D(\mathbb{P})$, $E \subset Q = \mathbb{P}(T^*\mathbb{P})$ be the complement of the largest open subset on which q^v is universally q^*F -analytic. Then, $SS(F)$ exists, and is the closed conical subset of $T^*\mathbb{P}$ s.t.

$$B(SS(F)) = \text{supp}(F).$$

$$\mathbb{P}(SS(F)) = E.$$

Proof. By Prop (2) $\Rightarrow$ (4), F is micro supp. on C^+ .

$\Rightarrow F$ is micro supp. on $SS(F) = C^+|_B$. $(F|_{\mathbb{P} \setminus B})$ is micro supp. on $\emptyset$.

Smallest: If F is micro supp. on $C' \subset T^*\mathbb{P}$,

$$\text{by prop (1)} \Rightarrow (2), \mathbb{P}(C') \supset E = \mathbb{P}(SS(F))$$

$$\text{Also } B(C') \supset \text{supp}(F) = B(SS(F))$$

$$\Rightarrow C' \supset SS(F). \quad \square$$

Coda ④ Beilinson further proved that, if X is connected/ k of dim n ,

$0 \neq F \in D(X)$, $SS(F)$ has pure dimension n . (holonomic)

But they are not Lagrangian in general.

When X is a smooth surface/ k , Deligne proved that every closed conical subset ^{irred.}

$C \subset T^*X$ of dimension 2 can arise as an irreducible component of some $SS(F)$

② Abbes - Saïto: Holonomic $\Rightarrow$ constructible.

